Computation of Conformal Mappings

Final Report on Schwarz-Christoffel Integrals and Numerical Methods

Final Project Report submitted to

Indian Institute of Technology Kharagpur

In partial fulfilment of the requirements for the award of the degree of

B.S.(Hons.) in Mathematics and Computing

By

Ravi Kumar

22MA10047

Under the guidance of

Prof. Bappaditya Bhowmik

DEPARTMENT OF MATHEMATICS
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

April 2026

DECLARATION

I certify that

a. the work contained in this report is original and has been done by me under the guidance of my supervisor(s).

b. the work has not been submitted to any other Institute for any degree or diploma.

c. I have followed the guidelines provided by the Institute in preparing the report.

d. I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

e. Whenever I have used material from other sources, I have cited the source in the report and listed it in the references.

Signature of the Student

CERTIFICATE

This is to certify that the report entitled, "**Computation of Conformal Mappings: Final Report on Schwarz-Christoffel Integrals and Numerical Methods**" submitted by **Mr. Ravi Kumar** to Indian Institute of Technology, Kharagpur, India, is a record of bonafide project work carried out by him under my supervision and guidance and is worthy of consideration for the award of the degree of B.S.(Hons.) in Mathematics and Computing of the Institute.

Prof. Bappaditya Bhowmik
Department of Mathematics
Supervisor

Date: April 2026

Abstract

This report studies the computational side of conformal mapping, with emphasis on Schwarz-Christoffel integrals and the numerical parameter problem. The first phase of the project introduced conformal maps, analytic functions, and the Schwarz-Christoffel formula as an explicit construction for polygonal domains. The final phase develops the problem further: given a target polygon, how are its prevertices and constants determined, how are singular integrals evaluated, and how do extremal principles provide alternate numerical constructions? The report discusses Newton iteration for side-ratio equations, singularity removal, disk-based Schwarz-Christoffel algorithms, variational methods, and the Bergman kernel. All diagrams in the technical body are original schematic illustrations prepared for this final report.

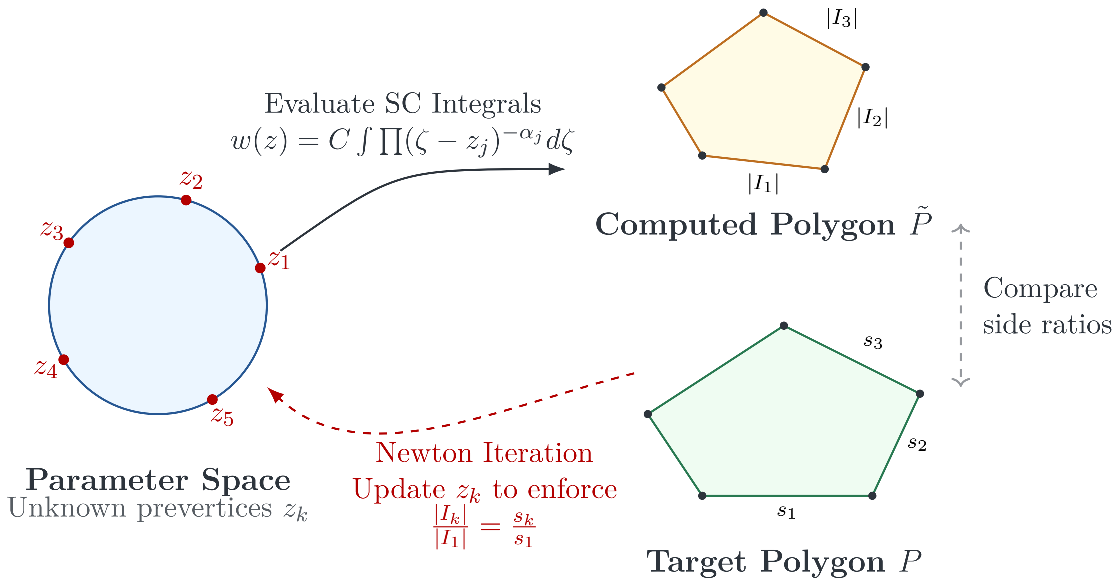

Figure 1: The numerical parameter problem: computing prevertices z_k such that the side lengths $|I_k|$ of the Schwarz-Christoffel map match the target side lengths s_k.

Contents

1 Introduction

Conformal mapping transforms one planar domain into another while preserving local angles. If $w = f(z)$ is analytic and $f'(z) \neq 0$, then infinitesimal figures are rotated and scaled but not sheared. This makes conformal mapping particularly useful for two-dimensional potential problems, where the real and imaginary parts of analytic functions solve Laplace's equation.

The earlier part of this BTP established the basic framework: analytic functions, Cauchy–Riemann equations, the Riemann mapping theorem, bilinear transformations, and the Schwarz-Christoffel formula. This final report uses that foundation but shifts the focus from theory to computation. The main concern is the numerical construction of conformal maps for polygonal domains.

A conformal map is locally equivalent to a rotation followed by a scaling. The scaling factor at z is $|f'(z)|$, while the local rotation is $\arg f'(z)$.

2 Schwarz-Christoffel Mapping Revisited

The Schwarz-Christoffel transformation maps a canonical domain, usually the upper half-plane or the unit disk, onto a polygonal domain. It is one of the few general formulas in conformal mapping that converts boundary geometry directly into an analytic expression. The key idea is simple: the real axis in the preimage plane is mapped to the polygon boundary, and each prevertex produces exactly the turn required at the corresponding polygon vertex.

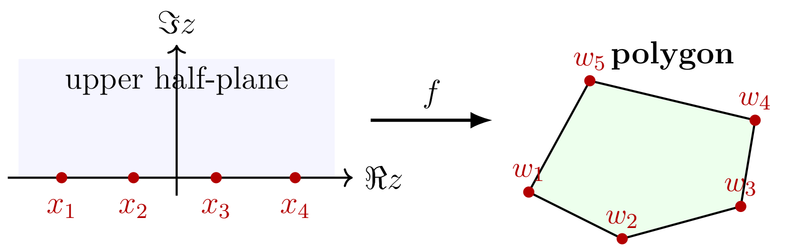

Figure 2: Boundary correspondence: real prevertices x_k map to polygon vertices w_k.

2.1 Formula and Angle Data

Let $x_1 < x_2 < \cdots < x_n$ be prevertices on the real axis and let $w_1, \ldots, w_n$ be the vertices of the polygon. If θ_k is the interior angle at w_k, define

$$\alpha_k = 1 - \frac{\theta_k}{\pi}.$$

The differential form of the upper half-plane Schwarz-Christoffel map is

$$\frac{dw}{dz} = C_1 \prod_{k=1}^{n} (z - x_k)^{-\alpha_k}.$$

Integrating gives the mapping function

$$w(z) = C_1 \int^z \prod_{k=1}^{n} (\zeta - x_k)^{-\alpha_k}\, d\zeta + C_2.$$

Here C_1 fixes scale and rotation, while C_2 fixes translation.

For a closed polygon,

$$\theta_k = (1 - \alpha_k)\pi, \qquad \sum_{k=1}^{n} \alpha_k = 2.$$

The sum condition is the analytic version of the fact that a closed polygon makes one full turn.

2.2 Argument Change at a Vertex

The factor $(z - x_k)^{-\alpha_k}$ is responsible for the corner at w_k. As z moves along the real axis and crosses x_k through the upper half-plane, the argument of $z - x_k$ changes by $-\pi$. Therefore

$$\Delta \arg(z - x_k)^{-\alpha_k} = -\alpha_k(-\pi) = \alpha_k \pi.$$

This is precisely the exterior turning angle required at the polygon vertex. The product of all such factors places all the required turns into the derivative dw/dz.

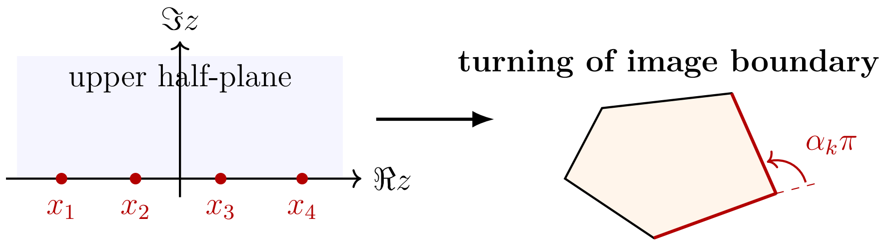

Figure 3: Each factor $(z - x_k)^{-\alpha_k}$ produces the required boundary turn at a polygon vertex.

2.3 Construction Procedure

The practical construction of an SC map follows four steps:

1. List the polygon vertices $w_1, \ldots, w_n$ in counterclockwise order.
2. Compute the angle parameters $\alpha_k = 1 - \theta_k/\pi$.
3. Fix three prevertices using Möbius normalization and solve for the remaining prevertices from side-length ratios.
4. Determine C_1 and C_2 from one side length and one vertex location.

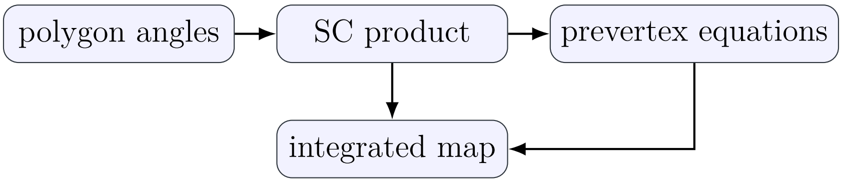

Figure 4: SC construction workflow from polygon geometry to mapping formula.

2.4 Example Mappings

2.4.1 General Polygon

For a general polygon with vertices $w_1, w_2, \ldots, w_n$, the real axis is divided into intervals by the prevertices $x_1, x_2, \ldots, x_n$. Each interval $[x_k, x_{k+1}]$ maps to one side of the polygon. Therefore, the side vector is given by

$$w_{k+1} - w_k = C_1 \int_{x_k}^{x_{k+1}} \prod_{j=1}^{n} (\zeta - x_j)^{-\alpha_j} \, d\zeta.$$

This equation is the geometric core of the Schwarz-Christoffel method: angles are determined by the exponents α_j, while side lengths are determined by the prevertex spacing.

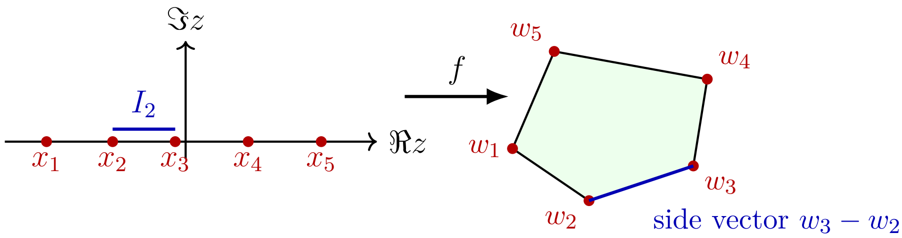

Figure 5: A prevertex interval in the upper half-plane maps to a corresponding polygon side.

2.4.2 Disk to Triangle

When the source domain is the unit disk instead of the upper half-plane, the Schwarz-Christoffel formula uses prevertices $z_k = e^{i\Theta_k}$ on the unit circle $|z| = 1$. For a triangle the formula is

$$w(z) = C_1 \int_0^z \prod_{k=1}^{3} (\zeta - z_k)^{-\alpha_k} \, d\zeta + C_2, \qquad z_k = e^{i\Theta_k}, \quad \alpha_k = 1 - \frac{\theta_k}{\pi}.$$

Three prevertices on the circle can again be fixed by a Möbius normalization; for a triangle all three are fixed, so no nonlinear parameter problem remains. A convenient standard choice is

$$z_1 = 1, \qquad z_2 = e^{2\pi i/3}, \qquad z_3 = e^{4\pi i/3}.$$

For an equilateral target triangle ($\alpha_k = 2/3$ for all k) the integrand is $(z^3 - 1)^{-2/3}$ up to a constant, and the integral again reduces to a hypergeometric type.

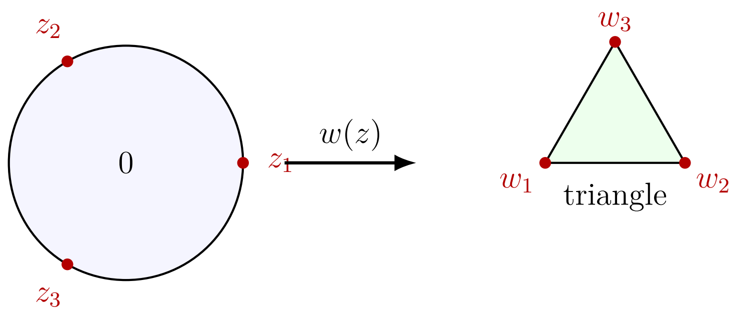

Figure 6: Disk to triangle: prevertices z_k placed symmetrically on $|z| = 1$ map to the vertices of the target triangle.

2.4.3 Polygon to Triangle via Composition

A direct conformal map from a general polygon P to a triangle T can be constructed by composing two SC maps:

1. Map P to the upper half-plane $\mathbb{H}$ via the *inverse* SC map $g : P \to \mathbb{H}$.
2. Map $\mathbb{H}$ to the triangle T via the SC map $f : \mathbb{H} \to T$.

The composed map $f \circ g : P \to T$ is then a conformal bijection. In formula,

$$\Phi = f \circ g : P \xrightarrow{g} \mathbb{H} \xrightarrow{f} T.$$

Neither f nor g has a compact closed form for a general polygon, but both can be evaluated numerically. The prevertex parameters for each stage are solved independently using the Newton iteration described in Section 4. After the two parameter problems are solved, function values of Φ at interior points are obtained by numerical quadrature along a path from a known boundary point.

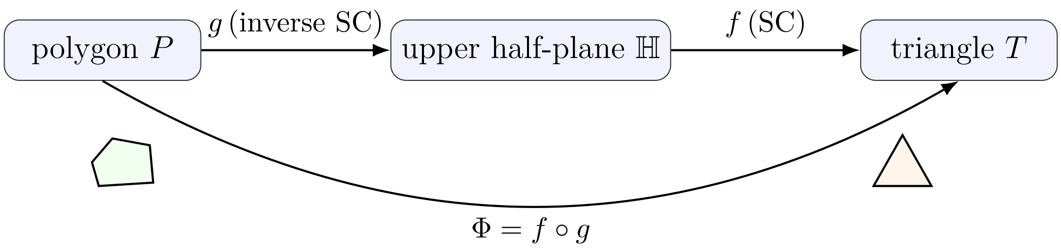

Figure 7: Polygon to triangle via two-step composition: inverse SC maps P to $\mathbb{H}$, then SC maps $\mathbb{H}$ to T.

2.5 Why the Parameter Problem Appears

The angles determine the exponents α_k, but they do not determine the side lengths. The relative side lengths depend on the prevertex spacing. Since real Möbius transformations preserve the upper half-plane, three prevertices may be fixed arbitrarily. The remaining $n - 3$ prevertices must be solved from side-ratio equations.

For adjacent vertices,

$$w_{k+1} - w_k = C_1 \int_{x_k}^{x_{k+1}} \prod_{j=1}^{n} (\zeta - x_j)^{-\alpha_j} \, d\zeta.$$

Taking ratios removes the unknown scale $|C_1|$, leaving nonlinear equations for the prevertices.

> The practical obstacle in Schwarz-Christoffel mapping is not the formula itself; it is the coupled numerical determination of prevertices and the accurate evaluation of singular integrals.

3 Schwarz-Christoffel Integrals

3.1 Side Integrals

The integrals between consecutive prevertices are called Schwarz-Christoffel integrals. A typical side integral is

$$I_k = \int_{x_k}^{x_{k+1}} \prod_{j=1}^{n} (\zeta - x_j)^{-\alpha_j} \, d\zeta.$$

The endpoints x_k and x_{k+1} are singular. For convex corners the singularities are integrable, but direct quadrature is inefficient unless the singular behavior is treated carefully.

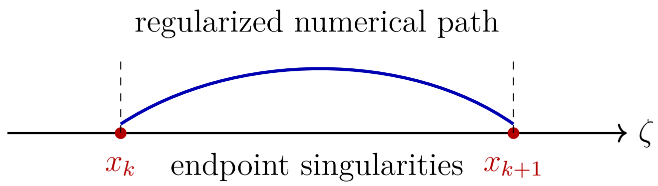

Figure 8: A side integral has singular behavior at both prevertices.

3.2 Kantorovich-Type Singularity Removal

Let the side integrand be written as

$$F(\zeta) = (\zeta - x_k)^{-\alpha_k} (x_{k+1} - \zeta)^{-\alpha_{k+1}} \Phi(\zeta),$$

where Φ is smooth on the interval. The idea is to subtract the leading endpoint singular terms, integrate those analytically, and then apply ordinary quadrature to the smooth remainder:

$$F(\zeta) = (\zeta - x_k)^{-\alpha_k} A_k + (x_{k+1} - \zeta)^{-\alpha_{k+1}} B_{k+1} + R(\zeta).$$

The constants A_k and B_{k+1} are chosen from local endpoint expansions. This improves numerical stability because $R(\zeta)$ is much less singular than $F(\zeta)$.

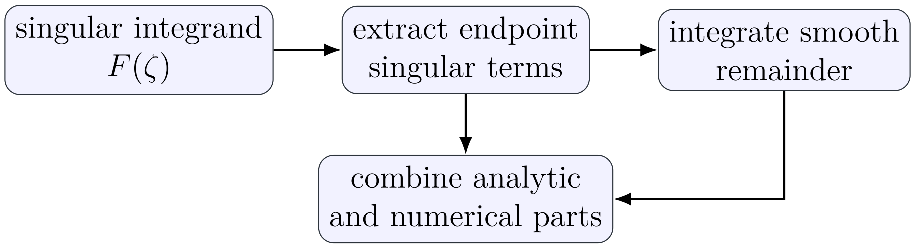

Figure 9: Singularity-removal workflow for evaluating SC side integrals.

4 Newton Method for the Parameter Problem

For an n-sided polygon, the side lengths are controlled by the spacing of the prevertices. A real Möbius transformation maps the upper half-plane onto itself and can send any three real boundary points to any other three real boundary points. Therefore three prevertices may be fixed as a normalization, leaving $n-3$ essential real unknowns. A common notation is

$$X = (x_3, x_4, \ldots, x_{n-1}).$$

Let the desired side-length ratios be

$$\lambda_j = \frac{|w_{j+1} - w_j|}{|w_2 - w_1|}, \qquad j = 2, \ldots, n-2.$$

If $I_j(X)$ denotes the SC integral over the j-th side interval, the parameter equations are

$$I_j(X) = \lambda_j I_1(X), \qquad j = 2, \ldots, n-2.$$

It is convenient to write these as residual equations

$$F_j(X) = I_j(X) - \lambda_j I_1(X), \qquad j = 2, \ldots, n-2.$$

The goal is to solve $F(X) = 0$.

4.1 Linearization and Jacobian

At an approximate solution $X^{(m)}$, Newton's method linearizes the residual:

$$F(X^{(m)} + h) \approx F(X^{(m)}) + J(X^{(m)})h.$$

The correction $h^{(m)}$ is obtained from

$$J(X^{(m)})h^{(m)} = -F(X^{(m)}), \qquad X^{(m+1)} = X^{(m)} + h^{(m)}.$$

The entries of the Jacobian are

$$J_{j\nu} = \frac{\partial F_j}{\partial x_\nu} = \frac{\partial I_j}{\partial x_\nu} - \lambda_j \frac{\partial I_1}{\partial x_\nu}, \qquad \nu = 3, \ldots, n-1.$$

In practice these derivatives may be obtained analytically by differentiating the integrand with respect to the prevertices, or numerically by finite differences. Analytical derivatives are more accurate but require careful handling near singular endpoints; finite differences are simpler but can become unstable when prevertices are close together.

4.2 Iteration and Stopping Criteria

The Newton iteration is stopped when both the residual and the update are small, for example when

$$\|F(X^{(m)})\| < \varepsilon_F \quad \text{and} \quad \|h^{(m)}\| < \varepsilon_X.$$

The prevertices must remain ordered on the real axis. If a Newton step breaks the ordering or makes the residual larger, one uses a damped step

$$X^{(m+1)} = X^{(m)} + \eta h^{(m)}, \qquad 0 < \eta \leq 1,$$

where η is reduced until the step is acceptable.

Newton's method is locally fast but not globally safe. It needs a reasonable initial guess, accurate quadrature for each SC integral, and safeguards when prevertices become crowded or nearly collide.

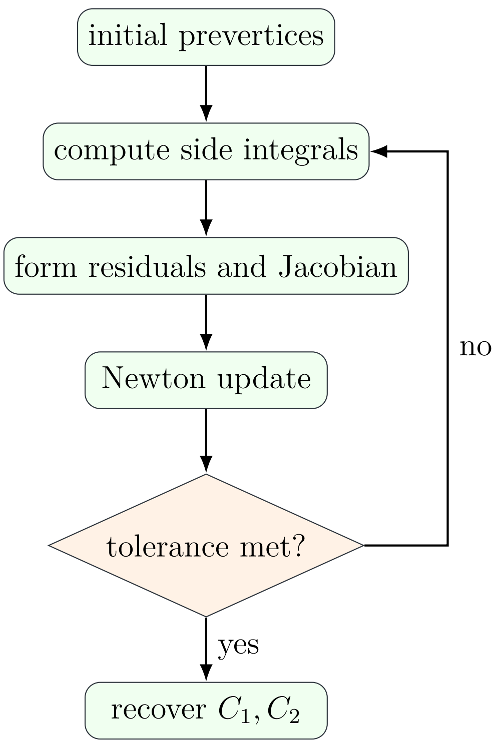

Figure 10: Newton iteration for matching side ratios in a polygon.

4.3 Quadrilateral Case

For a quadrilateral, three of the four prevertices can be fixed by normalization. The remaining prevertex can be represented by a single real parameter k, often chosen so that the four prevertices have a simple ordered form. The side-ratio condition becomes one scalar equation:

$$F(k) = I_4(k) - \lambda I_1(k) = 0.$$

The update is

$$k_{m+1} = k_m - \frac{F(k_m)}{F'(k_m)}.$$

This case is computationally important because many polygonal domains can be decomposed into quadrilateral-like blocks.

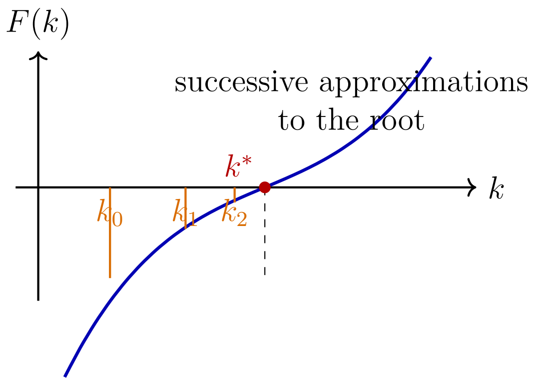

Figure 11: The quadrilateral parameter problem reduces to scalar root finding.

4.4 Representative Numerical Integral

A representative side integral in the Schwarz-Christoffel construction is

$$L_k = |w_{k+1} - w_k| = |C_1| \int_{x_k}^{x_{k+1}} \left| \prod_{j=1}^{n} (\zeta - x_j)^{-\alpha_j} \right| d\zeta.$$

The numerical task is to evaluate such integrals accurately. Since the integrand behaves like $|\zeta - x_k|^{-\alpha_k}$ near the left endpoint and $|\zeta - x_{k+1}|^{-\alpha_{k+1}}$ near the right endpoint, it possesses algebraic singularities at both ends of the integration interval. This is why quadrature accuracy and prevertex placement cannot be separated in practical computation.

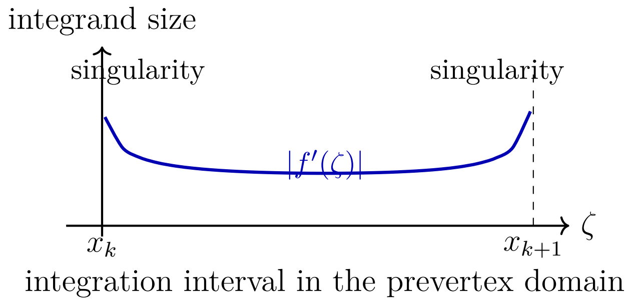

Figure 12: Repeated SC integral evaluations require quadrature adapted to endpoint singularities.

5 Disk Formulation and Trefethen-Type Algorithms

An alternative form maps the unit disk to the polygon. If $z_k = e^{i\Theta_k}$ are prevertices on the unit circle, then

$$w(z) = w_c + C \int_0^z \prod_{k=1}^{N} \left(\frac{z'}{z' - z_k} \right)^{-\alpha_k} dz'.$$

This formulation replaces real prevertex locations by angular unknowns Θ_k. Numerical algorithms combine nonlinear updates with quadrature rules adapted to algebraic endpoint singularities.

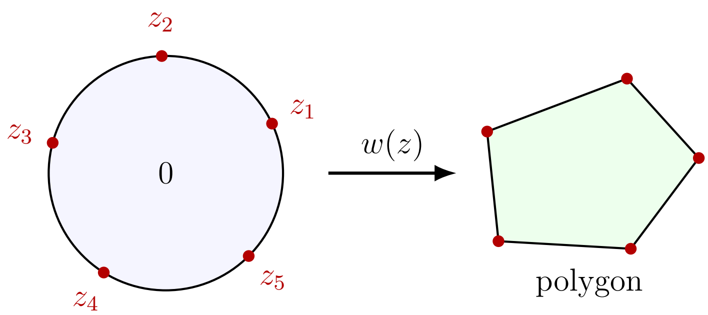

Figure 13: Unit-disk SC formulation with prevertices on $|z| = 1$.

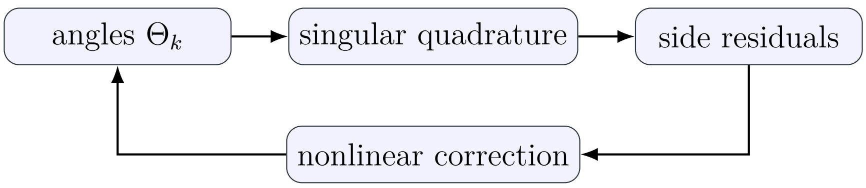

Figure 14: Coupling between quadrature and nonlinear parameter updates in disk-based algorithms.

6 Extremal Principles

An important approach to constructing conformal maps is through extremal principles. The idea, due to Gaier, is to characterize the conformal mapping of a simply connected region D onto a disk as the solution of a minimization problem in a Hilbert space of analytic functions. Once the minimizer is found, the map follows by integration.

6.1 Minimum Area Problem

The conformal mapping of a simply connected domain D onto a disk can be characterized through a variational principle known as the *Bieberbach minimizing principle*. Rather than specifying the boundary correspondence explicitly, this approach identifies the derivative of the Riemann mapping function as the unique solution to a norm-minimization problem in $L^2(D)$.

6.1.1 Function Classes and the Minimization Problem

Fix a base point $a \in D$. Define two classes of square-integrable analytic functions:

$$K_1(D) = \Big\{ f \in L^2(D) : f \text{ is analytic, } f(a) = 1 \Big\}, \qquad K_0(D) = \Big\{ f \in L^2(D) : f \text{ is analytic, } f(a) = 0 \Big\}.$$

Note that the classes $K_1(D)$ and $K_0(D)$ represent, respectively, a closed convex subset and a closed subspace of $L^2(D)$. Let the function

$$w = f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n, \qquad |z-a| < R,$$

which is regular in D, map D conformally onto the disk $B(0,R)$ in the w-plane. Without loss of generality, we will sometimes take the point a as the origin.

We will designate the minimum area problem as Problem I, defined as follows:

Problem I: In the class K_1, minimize the integral

$$I = \iint_D |f'(z)|^2 \, dS_z.$$

The Riemann mapping theorem guarantees the existence and uniqueness of the solution of this extremal problem.

Theorem (minimum area problem): Problem I has a unique solution $f_0(z) = f'(z)$. The minimum is πR^2.

Proof. Let $f(z)$ be a mapping function as in the power series above, normalized such that $f(a) = 0$ and $f'(a) = 1$. Then the area of the image of D is:

$$\begin{aligned} \operatorname{area}(f(D)) = \iint_D |f'(z)|^2 \, dS_z &= \int_0^R \int_0^{2\pi} \left| f'(re^{i\theta}) \right|^2 r \, dr \, d\theta \\ &= \pi R^2 |a_1|^2 + \pi \sum_{n=2}^{\infty} n \, |a_n|^2 R^{2n}. \end{aligned}$$

The above result implies that for a mapping function regular in the disk and normalized such that $f'(a) = a_1$, the area of the mapped region is always $\geq \pi R^2 |a_1|^2$. It is exactly equal to this value if the map is linear. For the normalized case $a_1 = 1$, the minimum area is πR^2, and the unique solution to the minimization problem is $f_0(z) = f'(z)$.

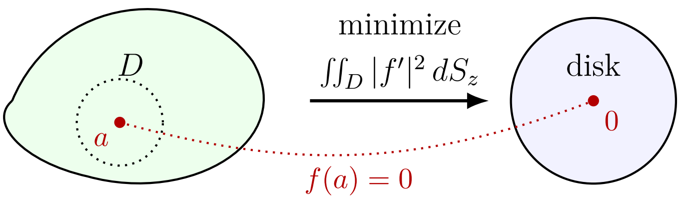

Figure 15: The Bieberbach minimizing principle: the derivative of the Riemann map uniquely minimizes the area integral over $K_1(D)$.

Theorem (Orthogonality). The minimizer f_0 is orthogonal in $L^2(D)$ to every $g \in K_0(D)$:

$$\langle f_0,\, g\rangle = \iint_D f_0(z)\, g(z)\, dS_z = 0.$$

Proof. For any $g \in K_0(D)$ and $\varepsilon > 0$, the function $f_0 + \varepsilon g$ lies in $K_1(D)$ since its value at a is still 1. By minimality:

$$\|f_0\|^2 \le \|f_0 + \varepsilon g\|^2 = \|f_0\|^2 + 2\varepsilon\, \Re\langle f_0, g\rangle + \varepsilon^2 \|g\|^2.$$

Dividing by ε and letting $\varepsilon \to 0^+$ forces $\Re\langle f_0, g\rangle \ge 0$. Repeating with $-\varepsilon$ gives the reverse inequality, and replacing g by ig settles the imaginary part. Hence $\langle f_0, g\rangle = 0$.

6.1.2 Connection to the Bergman Kernel

Substituting $g = f - f(a)$ (which belongs to $K_0(D)$) into the orthogonality relation gives, for every $f \in L^2(D)$:

$$\iint_D f_0(z)\, \overline{f(z)}\, dS_z = f(a) \cdot \|f_0\|^2.$$

Defining the *Bergman kernel* $K(z, a) = f_0(z)/\|f_0\|^2$, this becomes the reproducing property

$$\iint_D K(z, a)\, \overline{f(z)}\, dS_z = f(a), \qquad \text{for all } f \in L^2(D).$$

Setting $K(a, a) = 1/\|f_0\|^2$, the minimizer and the conformal map are then recovered as:

$$f_0(z) = \frac{K(z, a)}{K(a, a)}, \qquad f(z) = \frac{1}{K(a, a)} \iint_D K(z, a)\, dS_z.$$

Since $f_0 = f'$, the conformal mapping function itself is obtained by integrating f_0 from the base point a. In practice, $K(z, a)$ is not known analytically, but it can be approximated by orthonormalizing a polynomial basis in $L^2(D)$, which reduces the entire mapping problem to finite-dimensional linear algebra.

The Bieberbach principle converts the conformal mapping problem into a norm minimization over $K_1(D)$. The boundary of D need not be parametrized; only integrals over the domain interior are required to build the Bergman kernel and recover the map.

6.2 Bergman Kernel Method

The Bergman kernel method approximates $K(z, a)$ through a complete orthonormal system $\{\varphi_j^*(z)\}$ in the analytic $L^2(D)$ space:

$$K(z, a) = \sum_{j=1}^{\infty} \overline{\varphi_j^*(a)}\, \varphi_j^*(z).$$

In computation, the series is truncated after orthonormalizing a finite basis, often by Gram–Schmidt or an equivalent matrix factorization. The result is an approximate kernel $K_N(z,a)$, then an approximate minimal function

$$f_{0,N}(z) = \frac{K_N(z,a)}{K_N(a,a)}.$$

This approach emphasizes approximation of the reproducing kernel first and then recovers the map from the kernel representation.

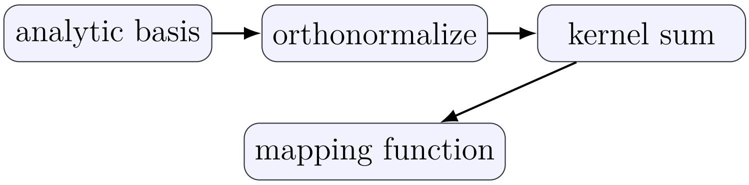

Figure 16: Bergman kernel method: build an orthonormal basis, approximate the kernel, then recover the map.

7 Applications

7.1 Potential Theory

Electrostatic potential, steady heat conduction, and ideal incompressible flow all lead to Laplace-type equations in two dimensions. A conformal map can move the problem from a complicated geometry to a simple canonical domain, where boundary data are easier to impose.

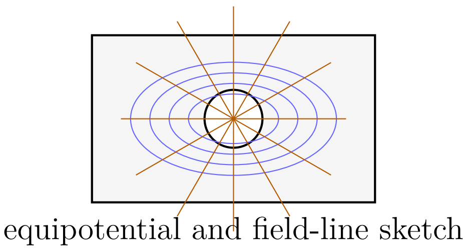

Figure 17: Conformal maps transfer potential fields between simple and complicated domains.

7.2 Fluid Flow

For irrotational incompressible flow, the complex potential $F = \phi + i\psi$ combines velocity potential and stream function. Streamlines and equipotential curves remain orthogonal under conformal maps, allowing flow around difficult boundaries to be studied through a simpler mapped geometry.

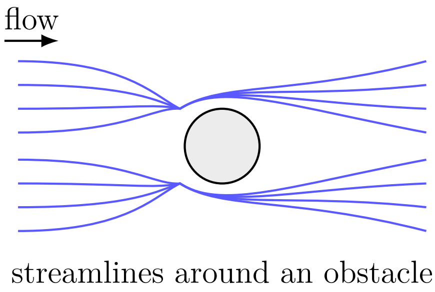

Figure 18: Potential-flow patterns can be generated by mapping canonical flows.

7.3 Computational Grid Generation

Boundary-fitted orthogonal grids improve numerical simulation near complicated boundaries. A conformal map naturally creates orthogonal coordinate curves, making it useful for mesh generation in polygonal regions.

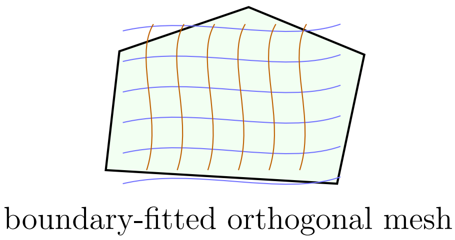

Figure 19: Conformal grids preserve orthogonality while adapting to boundary geometry.

8 Conclusion

This final report has focused on the computational issues that arise after the Schwarz-Christoffel formula is written down. The formula gives the correct angular structure of a polygonal map, but the actual construction requires solving a nonlinear parameter problem and evaluating singular integrals accurately. Newton iteration, singularity subtraction, disk-based algorithms, local approximations, and modulus-based approaches provide different ways to address this task.

The report also described extremal methods, where conformal maps are characterized through the minimum area principle, with the Bergman kernel providing the core theoretical representation. These approaches complement the Schwarz-Christoffel construction and show that conformal mapping is both an analytic and computational subject.

The practical relevance remains broad: potential theory, ideal fluid flow, heat transfer, and numerical grid generation all benefit from transforming complex domains into simpler canonical ones. For polygonal regions, Schwarz-Christoffel mapping remains one of the most direct and geometrically transparent tools for doing so.

Symbols and References

Symbol	Meaning
$z = x + iy$	Source-plane complex variable
$w = u + iv$	Image-plane complex variable
$f(z)$	Conformal mapping function
D	Physical domain
Γ	Boundary of D
x_k	Real prevertex in the upper-half-plane formulation
$z_k = e^{i\Theta_k}$	Unit-circle prevertex in the disk formulation
w_k	Polygon vertex
θ_k	Interior angle at w_k
α_k	Exterior-angle parameter, $1 - \theta_k/\pi$
C_1, C_2	Scaling/rotation and translation constants
I_k	Schwarz-Christoffel side integral
$K(z, a)$	Bergman kernel

[1] Prem K. Kythe, *Handbook of Conformal Mappings and Applications.* CRC Press, 2019.

[2] Lloyd N. Trefethen, *Numerical Conformal Mapping.*